Supervised Learning Assignment #03

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Introduction

This report presents a robust object detection approach for identifying an elephant within a cluttered desk image using Scale-Invariant Feature Transform (SIFT). The method combines local feature extraction, descriptor matching via FLANN, filtering through Lowe's ratio test, and geometric verification using RANSAC-based homography estimation. A polygonal outline derived from annotated template points is projected onto the target image to evaluate localization accuracy. By tuning key parameters including SIFT detector settings, matching thresholds, and RANSAC tolerance, the detection performance is significantly improved. Although perfect detection is not achievable due to noise and clutter, the approach demonstrates reliable object localization.

Methodology

2.1. SIFT Feature Detection and Description

SIFT (Scale-Invariant Feature Transform) extracts distinctive key points and descriptors that are invariant to scale and rotation. Tuned parameters include (e.g. **increasing number of features** captures more keypoints on the elephant, **decreasing Contrast threshold** detects weaker but useful features**, edge threshold adjusted** reduces unstable edge responsesThese improvements ensured better coverage of the elephant's structure.

2.2. Feature Matching using FLANN:

A FLANN-based matcher was used for efficient nearest-neighbor search between descriptors. FLANN improves scalability compared to brute-force matching, especially when dealing with a large number of descriptors.

2.3. Lowe's Ratio Test:

to remove ambiguous matches, I applied Lowe's ratio test. From experimentation, lower values are fewer but more reliable matches whereas higher values are more matches but more noise. This step is crucial for eliminating incorrect correspondences.

2.4. Homography Estimation using RANSAC:

After obtaining good matches, a homography matrix is estimated.

2.5. Polygon Projection (Object Localization)

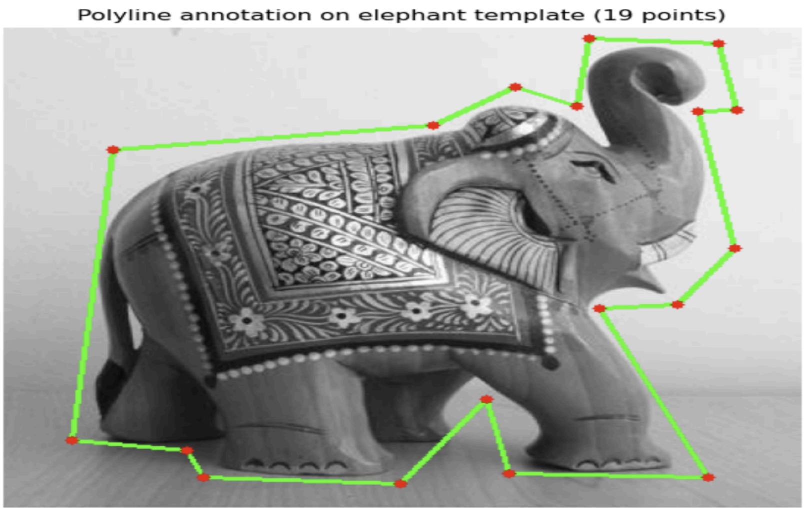
Using the annotation tool, approximately **19 points** were defined on the elephant template.

1. Transform points using homography
2. Project them onto the cluttered image
3. Draw a polygon connecting the transformed points

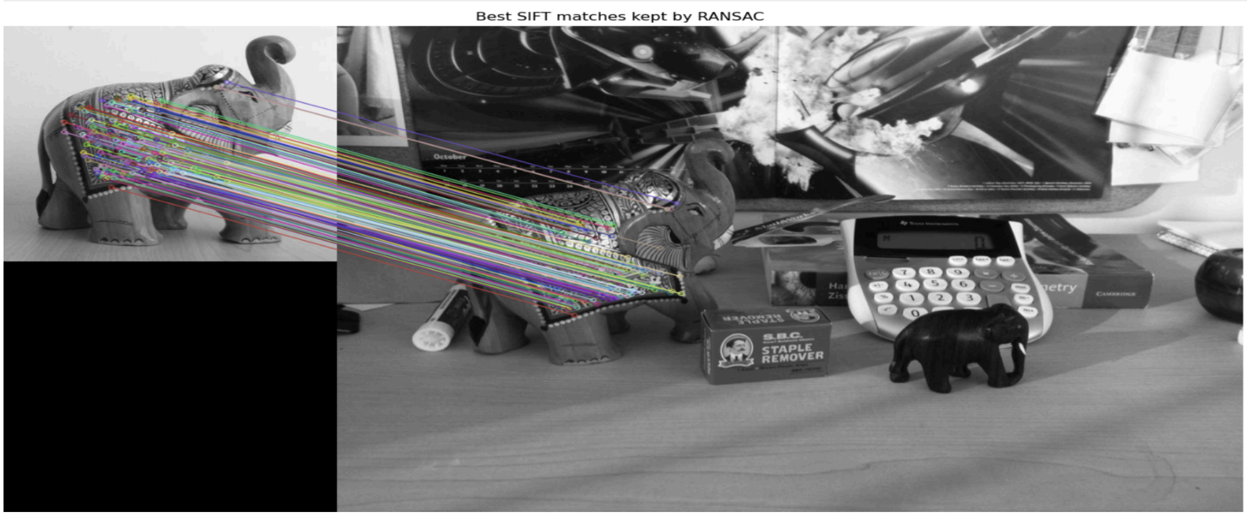
This provides a visual representation of how accurately the elephant was detected.

Results

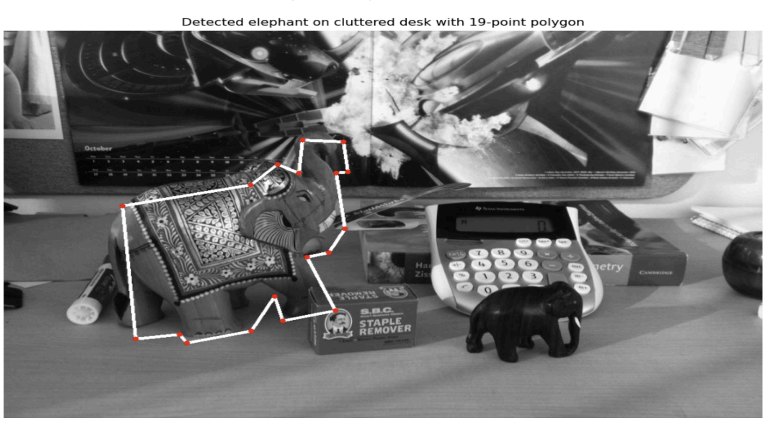
1. Annotated image of an elephant and the points extracted from json file export from the annotation tool




```
Number of polygon points extracted from the Json file: 19
[[ 33. 322.]
 [ 53.  95.]
 [209.  76.]
 [249.  46.]
 [279.  61.]
 [285.   8.]
 [348.  12.]
 [357.  64.]
 [338.  65.]
 [356. 172.]
 [328. 216.]
 [290. 219.]
 [343. 351.]
 [246. 348.]
 [235. 290.]
 [193. 356.]
 [ 97. 351.]
 [ 89. 330.]
 [ 33. 322.]]
```



2. Feature Matching Using SIFT and FLANN




```
Detected polygon points on clutteredDesk.jpg:
[[180.46 543.24]
 [162.13 309.29]
 [336.5  271.79]
 [374.37 236.42]
 [402.86 250.73]
 [407.66 192.94]
 [463.81 195.99]
 [469.91 249.06]
 [454.26 251.18]
 [466.34 349.36]
 [444.1  390.69]
 [414.04 399.  ]
 [453.41 492.61]
 [380.88 510.67]
 [370.21 467.84]
 [337.68 528.94]
 [250.22 549.56]
 [239.93 534.56]
 [180.46 543.24]]
```

3. Elephant Detection and detected polygon points on clutter image

Key tasks done and parameters used summary from code

```
Summary
==========
Object searched            : Elephant
Polyline points used       : 19
Template keypoints         : 2671
Scene keypoints            : 5000
Good matches after ratio   : 215
RANSAC inliers             : 176
Inlier ratio               : 81.86%
Polygon area               : 72688.38
Polygon points inside image : 100.00%

Best parameter set used:
  - nfeatures: 5000
  - nOctaveLayers: 5
  - contrastThreshold: 0.02
  - edgeThreshold: 12
  - sigma: 1.4
  - trees: 8
  - checks: 80
  - ratio_thresh: 0.75
  - ransac_tol: 5.0
  - min_matches: 12
  - min_template_kp: 20
  - min_scene_kp: 50

The detection is not perfect. The object is partially small, surrounded by clutter, and the template-to-scene appearance is not identical. Even so, the
tuned SIFT + FLANN + RANSAC pipeline gives a good practical localization of the elephant.
```

Conclusion

SIFT-based feature matching combined with RANSAC is an effective approach for detecting objects in cluttered environments.

Key points:

- Careful parameter tuning significantly improves performance
- Feature-based methods outperform naive template matching
- Homography enables accurate object localization

However, perfect detection remains challenging due to noise and environmental complexity.